Lengths and ratios

L4-GE2 Level 4 Olympiad · Geometry

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Power of a point, radical axes, the ratio lemma with Ceva and Menelaus, and mass points, for AIME computation and BMO proofs.
By the end you can use signed power of a point, radical axes and radical centres in proofs and computations; turn angle data into side ratios with the ratio lemma and trig Ceva; read off ratios and areas in cevian configurations with mass points.
Prerequisites: Power of a point and cyclic quadrilaterals (L3-GE1), Trigonometry and coordinates (L3-GE2), Olympiad angle chasing (L4-GE1). **Contests:** BMO1, MOG, AIME, USAJMO. **Time:** about 120 minutes.

1 Lesson

1.1 Power of a point

Theorem 1 (power of a point).
Let ω have centre O and radius r. If a line through P meets ω at A and B, then

$$\overline{PA} \cdot \overline{PB} = OP^2 - r^2,$$

where bars denote signed lengths along the line. This is the *power* $\operatorname{pow}(P)$, which is negative inside ω and equals PT^2 for a tangent PT.

Proof. With M the midpoint of AB, so that $OM \perp AB$, $\overline{PA} \cdot \overline{PB} = (\overline{PM} + \overline{MA})(\overline{PM} - \overline{MA}) = PM^2 - MA^2 = (OP^2 - OM^2) - (r^2 - OM^2)$. A tangent is the case $A = B = T$.

Theorem 2 (converse).
If lines AB and CD meet at P and $\overline{PA} \cdot \overline{PB} = \overline{PC} \cdot \overline{PD}$, then A, B, C, D are concyclic. If T is not on line AB and $PT^2 = \overline{PA} \cdot \overline{PB}$, then PT is tangent to circle ABT.

Proof. Line PC meets circle ABC again at D', and Theorem 1 gives $\overline{PC} \cdot \overline{PD'} = \overline{PA} \cdot \overline{PB} = \overline{PC} \cdot \overline{PD}$, so $D' = D$. For the tangent, line PT meets circle ABT again at T' with $\overline{PT} \cdot \overline{PT'} = PT^2$, so $T' = T$.

1.2 Radical axis and radical centre

Theorem 3 (radical axis).
For two circles with different centres, the points with equal power form a line perpendicular to the line of centres. If the circles meet at X and Y, the radical axis is line XY.
Theorem 4 (radical centre). The three radical axes of three circles with non-collinear centres pass through one point.

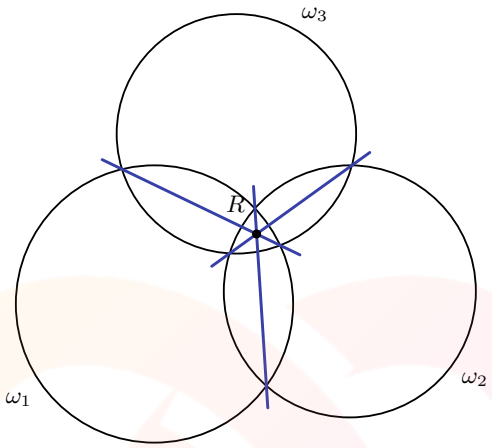
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Proof. With centres $(0,0)$ and $(d,0)$, equal power reads $x^2+y^2-r_1^2=(x-d)^2+y^2-r_2^2$, that is

$$x=\frac{d^2+r_1^2-r_2^2}{2d}.$$

A common point has power 0 for both circles, so the point lies on this line. For Theorem 4, two of the axes are perpendicular to non-parallel lines, so they meet at a point R, which has equal power for all three circles.



So three common chords of three circles are concurrent.

1.3 The ratio lemma, Ceva and Menelaus

Theorem 5 (ratio lemma).
If D lies on side BC of triangle ABC, then

$$\frac{BD}{DC}=\frac{AB\sin\angle BAD}{AC\sin\angle DAC}.$$

Proof. Triangles ABD and ADC have the same height from A, so

$$\frac{BD}{DC}=\frac{[ABD]}{[ADC]}=\frac{\frac{1}{2}AB\cdot AD\sin\angle BAD}{\frac{1}{2}AC\cdot AD\sin\angle DAC}$$

.

Theorem 6 (Ceva).
For D,E,F on sides BC,CA,AB, the lines AD,BE,CF are concurrent if and only if

$$\frac{BD}{DC}\cdot\frac{CE}{EA}\cdot\frac{AF}{FB}=1,$$

or equivalently (trig Ceva) $\frac{\sin\angle BAD}{\sin\angle DAC}\cdot\frac{\sin\angle CBE}{\sin\angle EBA}\cdot\frac{\sin\angle ACF}{\sin\angle FCB}=1$.

Proof. If the cevians meet at P, then

$$\frac{BD}{DC}=\frac{[ABD]-[PBD]}{[ADC]-[PDC]}=\frac{[ABP]}{[ACP]}$$

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, and the three such ratios multiply to 1. Conversely, let AD and BE meet at P and let CP meet AB at F'. Then $AF'/F'B = AF/FB$, so $F' = F$. Trig Ceva is Theorem 5 applied to each ratio, and the sides cancel.

Theorem 7 (Menelaus).
If a line meets lines BC, CA, AB at D, E, F, then the same product of unsigned ratios equals 1. Conversely, if the product is 1 and one or three of the points lie outside the sides, then D, E, F are collinear.

Proof. Let h_A, h_B, h_C be the distances from A, B, C to the line. By similar triangles $BD/DC = h_B/h_C$, and the product is $\frac{h_B}{h_C} \cdot \frac{h_C}{h_A} \cdot \frac{h_A}{h_B} = 1$. The converse goes as for Ceva.

1.4 Mass points

Theorem 8 (mass points).
Put masses $m_A, m_B, m_C > 0$ at the vertices and let $P = \frac{m_A A + m_B B + m_C C}{m_A + m_B + m_C}$. Then line AP meets BC at D with $BD : DC = m_C : m_B$ and $AP : PD = (m_B + m_C) : m_A$, and similarly at B and C. Moreover $[PBC] : [PCA] : [PAB] = m_A : m_B : m_C$.

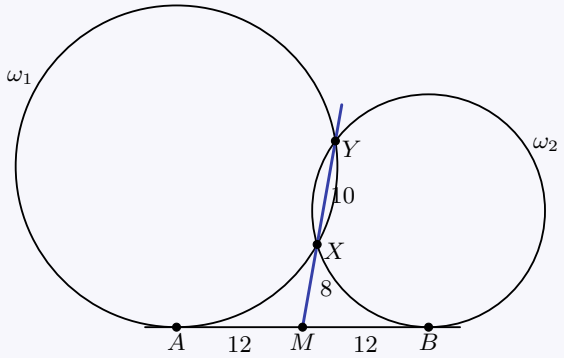
Proof. Group B with C. With $D = \frac{m_B B + m_C C}{m_B + m_C}$ we get $P = \frac{m_A A + (m_B + m_C) D}{m_A + m_B + m_C}$, which is the lever law on BC and on AD. For the areas, $[PBC] : [ABC] = PD : AD = m_A : (m_A + m_B + m_C)$.

Grouping is the whole method, since any set of the masses may be replaced by its total at its centre of mass. This also lets you split the mass at a vertex between two sides (Problem 5).

1.5 Worked examples

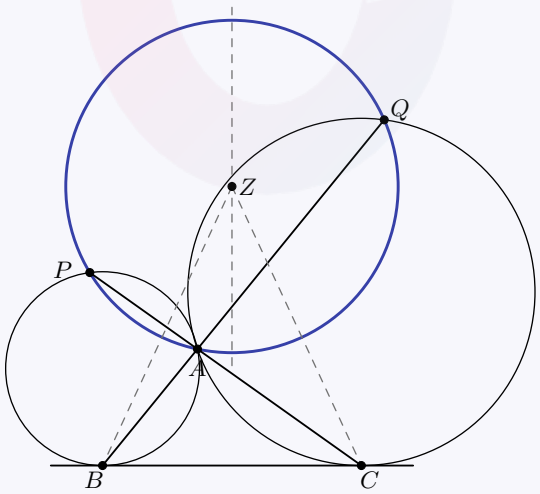
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Example 1.
Circles ω_1 and ω_2 meet at X and Y. A common tangent, nearer to X than to Y, touches ω_1 at A and ω_2 at B. Given $AB = 24$ and $XY = 10$, find the distance from X to the midpoint of AB.



Solution.
Let line XY meet AB at M. By Theorem 3, $MA^2 = \overline{MX} \cdot \overline{MY} = MB^2$, so M is the midpoint of AB and $MA = 12$. As M is outside both circles, X and Y lie on the same side of M. With $MX = x$ we get $x(x + 10) = 144$, so $x = 8$.
Remark. The common chord bisects every common tangent. With no radius given, powers beat coordinates.

Example 2.
In triangle ABC, the circle through A tangent to BC at B meets line AC again at P, and the circle through A tangent to BC at C meets line AB again at Q. Prove that the centre of circle APQ is equidistant from B and C.

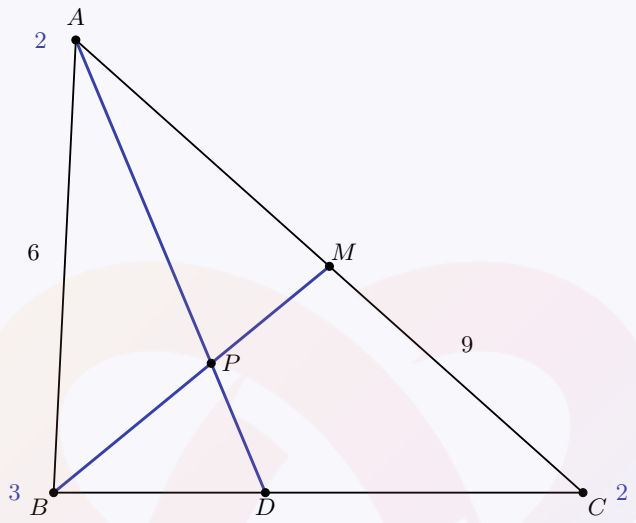


Solution.
Let Z and ρ be the centre and radius of circle APQ. As $ZB^2 = \rho^2 + \mathrm{pow}(B)$ and $ZC^2 = \rho^2 + \mathrm{pow}(C)$, we only need to show that B and C have equal power.
B lies on line AQ, so its power is $\overline{BQ} \cdot \overline{BA}$. This is also the power of B with respect to the circle through A and Q tangent to BC at C, which is BC^2. In the same way the power of C is $\overline{CP} \cdot \overline{CA} = CB^2$. ■

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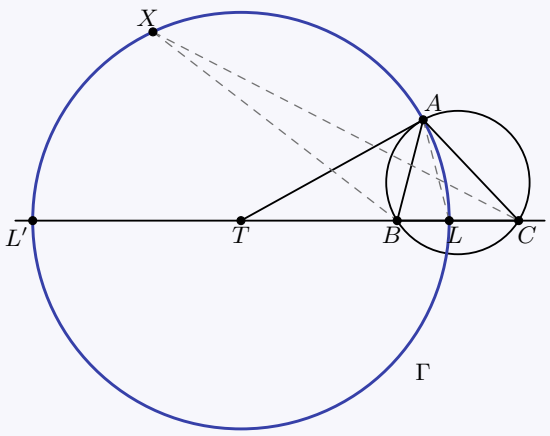
Remark. The identity $ZX^2 = \rho^2 + \mathrm{pow}(X)$ turns equal distances from a centre into equal powers, and signed lengths cover every position of P and Q.

Example 3.
Triangle ABC has area 21, $AB = 6$ and $AC = 9$. The bisector of angle A meets BC at D, the median from B meets AC at M, and AD meets BM at P. Find $AP : PD$, $BP : PM$ and the area of triangle APB.



Solution.
By the angle bisector theorem $BD : DC = 6 : 9$, so we need $m_C : m_B = 2 : 3$, and M is the midpoint of AC, so $m_A = m_C$. Take masses $2, 3, 2$ at A, B, C. Theorem 8 gives $AP : PD = 5 : 2$, $BP : PM = 4 : 3$ and $[APB] = \frac{2}{7} \cdot 21 = 6$.
Remark. Two ratios fix the masses, and then everything is read off, and BC is never needed.

Example 4.
Triangle ABC has $AB = 6$, $AC = 8$, $BC = 7$, and the tangent at A to its circumcircle meets line BC at T. Find TA, and prove that every point X of the circle Γ with centre T through A satisfies $XB : XC = 3 : 4$.



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Solution.
Since $AB < AC$, T lies beyond B. By the alternate segment theorem $\angle TAB = \angle ACB$, so triangles TAB and TCA are similar and

$$\frac{TA}{TC} = \frac{TB}{TA} = \frac{AB}{CA} = \frac{3}{4}.$$

Thus $TB = \frac{9}{16}TC$, and $TC - TB = 7$ gives $TC = 16$, $TB = 9$, $TA = 12$.
For X on Γ off line BC, $TX^2 = TA^2 = TB \cdot TC$, so TX is tangent to circle XBC (Theorem 2), and as before $XB : XC = TX : TC = 3 : 4$. The two points of Γ on line BC are checked directly. ■
Remark. Γ is the Apollonius circle of B, C for the ratio $3 : 4$. Its point L on segment BC is the foot of the bisector, so $TA = TL$, which is worth remembering.

Common traps.

- Mass points backwards, as in $BD : DC = m_C : m_B$, not $m_B : m_C$.
- Inside a circle, $PA \cdot PB = r^2 - OP^2$, so signed lengths avoid the case split.
- The converse of Menelaus needs one or three points outside the sides, while Ceva needs all three on the sides.

1.6 How to spot this in a contest

1. A common chord or common tangent suggests the radical axis. Three lines to be shown concurrent suggest the radical centre.
2. Angles at a vertex and a ratio on the opposite side suggest the ratio lemma.
3. Ratios on cevians, or areas cut off by cevians, suggest mass points.

2 Practice problems

The problems are roughly in order of difficulty, from ⋆ to ⋆⋆⋆⋆. Hints are in Section 4 and full solutions in Section 5.

Problem 1 ⋆ In triangle ABC, $AB = 6$ and $AC = 10$. The midpoint M of BC satisfies $\angle BAM = \angle BCA$. Find BC and AM.

Problem 2 ⋆⋆ In triangle ABC, the median from A meets the circumcircle again at A'. Prove that $AB \cdot BA' = AC \cdot CA'$.

Problem 3 ⋆⋆ From a point P outside a circle ω with centre O, the tangents touch ω at A and B, and M is the midpoint of AB. A chord CD of ω passes through M and does not lie on line OP. Prove that P, O, C, D are concyclic.

Problem 4 ⋆⋆⋆ Let ABC be a triangle without a right angle, with orthocentre H. Points D, E, F are chosen anywhere on the sides BC, CA, AB. Prove that H has the same power with respect to the three circles with diameters AD, BE and CF.

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Problem 5 ⋆⋆⋆ A line through the incentre I of triangle ABC meets the sides AB and AC at P and Q, both different from A. Prove that

$$CA \cdot \frac{PB}{PA} + AB \cdot \frac{QC}{QA} = BC.$$

Problem 6 ⋆⋆⋆⋆ Let ABC be a scalene triangle whose incircle touches BC, CA, AB at D, E, F. Line EF meets line BC at X, line FD meets line CA at Y, and line DE meets line AB at Z. Prove that the midpoints of DX, EY and FZ are collinear.

3 Contest ladder

Real past problems on the ideas of this unit, easiest first. Each source is a link to the official paper or the AoPS wiki; the note under it says where to start, not how to finish.

1. **AIME 1985 Problem 6** ⋆⋆
   Triangles with a common height have areas in the ratio of their bases, and two unknown areas need two equations.
2. **AIME 1992 Problem 13** ⋆⋆
   The vertex C moves on a fixed circle, as in Example 4. Where on that circle is the area largest?
3. **BMO1 2009/10 Problem 4** ⋆⋆
   Express everything through the two radii, and find the angle between DE and BC.
4. **AIME 1989 Problem 15** ⋆⋆⋆
   Mass points give every ratio on the cevians, then Stewart's theorem.
5. **BMO1 2007/08 Problem 5** ⋆⋆⋆
   Write each ratio in terms of the area weights of P, then use AM-GM.
6. **USAJMO 2012 Problem 1** ⋆⋆⋆
   Read each angle condition as a tangency, then compare powers of points.
7. **IMO 2000 Problem 1** ⋆⋆⋆
   Line NM bisects AB, as in Example 1, so M is the midpoint of PQ. Then show that EM is perpendicular to CD.
8. **BMO1 2025/26 Problem 4** ⋆⋆⋆
   Angles from the two tangencies give similar triangles with a common vertex at M.
9. **BMO1 2012/13 Problem 6** ⋆⋆⋆
   The tangencies give a pair of similar triangles sharing the vertex D.
10. **BMO1 2016/17 Problem 5** ⋆⋆⋆
    Turn equal areas into equal lengths, then use the power of the point Q.
11. **IMO 1995 Problem 1** ⋆⋆⋆⋆
    The power of P shows that B, C, M, N are concyclic. Then find a circle through A, M, N, D and use the radical centre.
12. **IMO 2009 Problem 2** ⋆⋆⋆⋆
    The tangency and the midline parallels give a pair of similar triangles. Rewrite the result as equal powers of P and Q with respect to the circumcircle.

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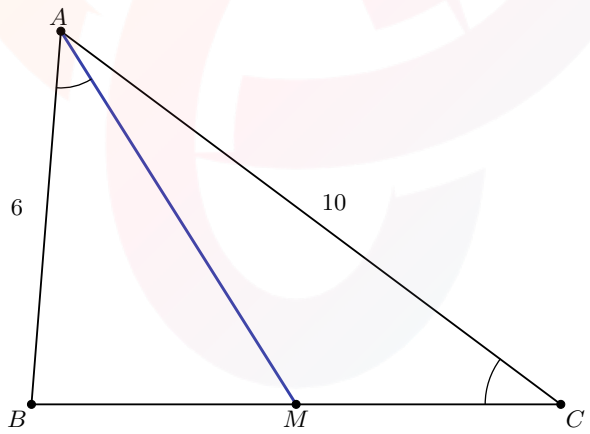
4 Hints

1. Look for two similar triangles sharing the angle at B.
2. Compare the areas of triangles ABA' and ACA' in two ways.
3. Find two different expressions for MA^2.
4. Besides A and D, name one more point of the circle with diameter AD, one that does not depend on D.
5. Masses BC, CA, AB at A, B, C balance at I. Now split the mass at A into one part for P and one for Q.
6. Menelaus gives $XB : XC$. Then look for two circles with respect to which the midpoint of DX has equal powers.

5 Solutions

Problem 1

In triangle ABC, $AB = 6$ and $AC = 10$. The midpoint M of BC satisfies $\angle BAM = \angle BCA$. Find BC and AM.



Triangles BAM and BCA share the angle at B, and $\angle BAM = \angle BCA$, so they are similar. Hence

$$\frac{BA}{BC} = \frac{BM}{BA} = \frac{AM}{CA}.$$

The first equality gives $BA^2 = BM \cdot BC = \frac{1}{2}BC^2$, so $BC = \sqrt{2}\,AB = 6\sqrt{2}$. Then $AM = CA \cdot \frac{BA}{BC} = \frac{10}{\sqrt{2}} = 5\sqrt{2}$.

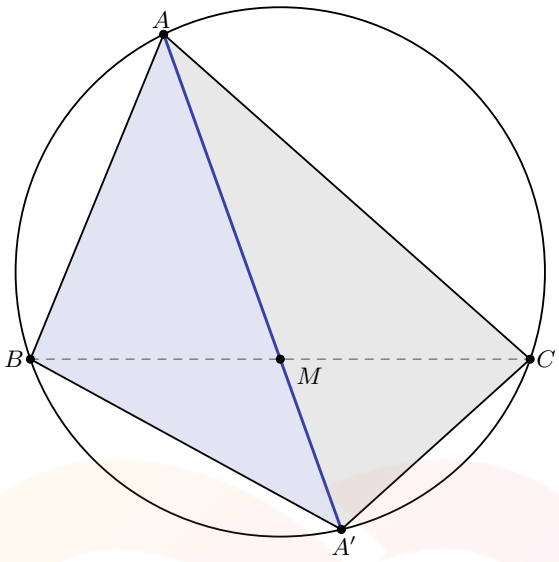
Conversely, sides 6, $6\sqrt{2}$, 10 do form a triangle, and $BA^2 = BM \cdot BC$ makes the two triangles similar, so the condition holds. In the language of this unit, BA is tangent to circle AMC and BA^2 is the power of B.

Answer: $BC = 6\sqrt{2}$, $AM = 5\sqrt{2}$

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Problem 2

In triangle ABC, the median from A meets the circumcircle again at A'. Prove that $AB \cdot BA' = AC \cdot CA'$.



Let M be the midpoint of BC. Since M lies on AA' and $BM = MC$, the points B and C are at the same distance from line AA', so $[ABA'] = [ACA']$.

The quadrilateral $ABA'C$ is cyclic with B and C on opposite sides of AA', so $\angle ABA' + \angle ACA' = 180^\circ$ and the two angles have equal sines. Therefore

$$\tfrac{1}{2}\, AB \cdot BA' \sin\angle ABA' = \tfrac{1}{2}\, AC \cdot CA' \sin\angle ACA'$$

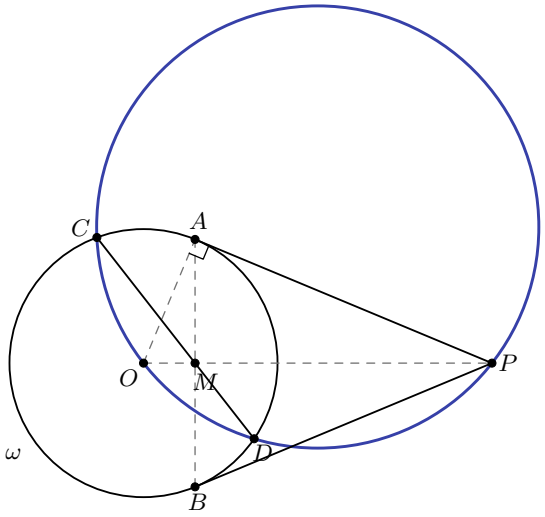
gives $AB \cdot BA' = AC \cdot CA'$. ■

Equivalently $A'B : A'C = AC : AB$, which is the ratio lemma for the cevian $A'M$ of triangle $BA'C$.

Problem 3

From a point P outside a circle ω with centre O, the tangents touch ω at A and B, and M is the midpoint of AB. A chord CD of ω passes through M and does not lie on line OP. Prove that P, O, C, D are concyclic.

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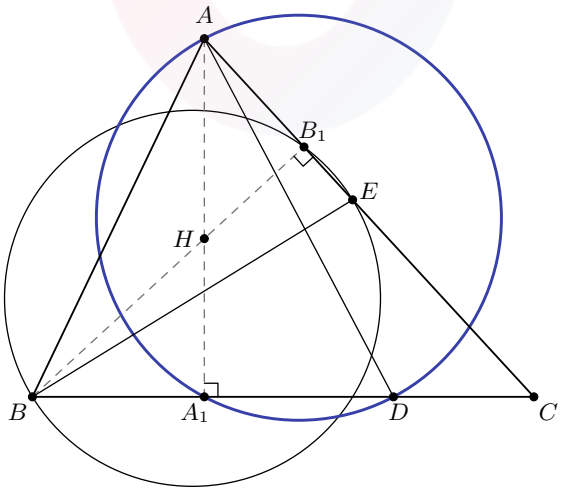


Since $PA = PB$ and $OA = OB$, line OP is the perpendicular bisector of AB, so this line passes through M and $AM \perp OP$. The radius OA is perpendicular to the tangent PA, so AM is the altitude to the hypotenuse of the right-angled triangle OAP. Hence M lies between O and P, and the similar triangles OMA and AMP give $MO \cdot MP = MA^2$.

The chords AB and CD meet at M, so Theorem 1 gives $MC \cdot MD = MA \cdot MB = MA^2$. Thus $MO \cdot MP = MC \cdot MD$, where M lies between O and P and between C and D, and the two lines are different. By Theorem 2, P, O, C, D are concyclic. ■

Problem 4

Let ABC be a triangle without a right angle, with orthocentre H. Points D, E, F are chosen anywhere on the sides BC, CA, AB. Prove that H has the same power with respect to the three circles with diameters AD, BE and CF.



Let AA_1, BB_1, CC_1 be the altitudes. Since $\angle AA_1D = 90°$, the foot A_1 lies on the circle with diameter AD (if $D = A_1$ then A_1 is an end of the diameter). So line AH meets this circle at A and A_1, and by Theorem 1 the power of H is $\overline{HA} \cdot \overline{HA_1}$. In the same way the other two powers are $\overline{HB} \cdot \overline{HB_1}$ and $\overline{HC} \cdot \overline{HC_1}$.

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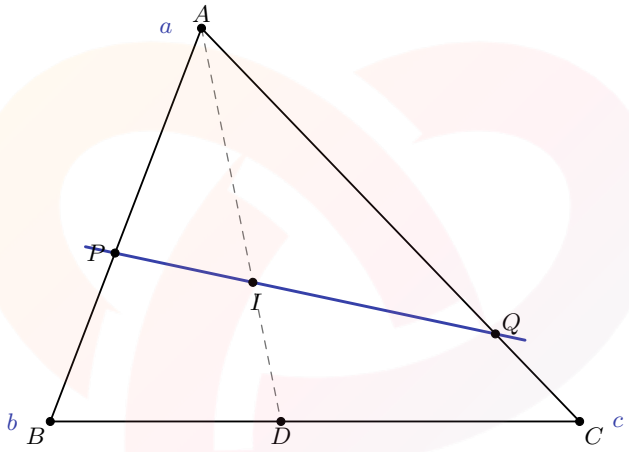
These products are equal. As $\angle AA_1B = \angle AB_1B = 90°$, the points A, B, A_1, B_1 lie on the circle with diameter AB, and Theorem 1 at H, for the chords AA_1 and BB_1, gives $\overline{HA} \cdot \overline{HA_1} = \overline{HB} \cdot \overline{HB_1}$. The circle with diameter BC gives the second equality in the same way. ■

So whenever the three centres are not collinear, H is the radical centre, wherever D, E, F are. Since the answer does not depend on D, the first thing to look for was a point of each circle that does not depend on D either.

Problem 5

A line through the incentre I of triangle ABC meets the sides AB and AC at P and Q, both different from A. Prove that

$$CA \cdot \frac{PB}{PA} + AB \cdot \frac{QC}{QA} = BC.$$



Write $a = BC$, $b = CA$, $c = AB$, and let AI meet BC at D. Masses a, b, c at A, B, C have centre of mass I, because by the angle bisector theorem $BD : DC = c : b$, so by Theorem 8 the centre of mass lies on AD, and in the same way on the bisector from B. Theorem 8 also gives $AI : ID = (b + c) : a$.

Now let $x = b \cdot \frac{PB}{PA}$ and $y = c \cdot \frac{QC}{QA}$. Then $PA : PB = b : x$, so P is the centre of mass of x at A and b at B, and likewise Q is the centre of mass of y at A and c at C. Let G be the centre of mass of $x + y$ at A, b at B and c at C. Grouping the masses as $\{x, b\}$ and $\{y, c\}$ shows that G lies on PQ. Grouping them as $\{x + y\}$ and $\{b, c\}$ shows that G lies on AD with $AG : GD = (b + c) : (x + y)$.

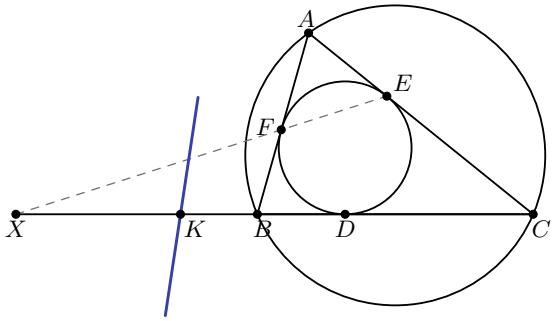
Line PQ is not line AD, since otherwise $P = Q = A$. So the two lines meet only at I, and $G = I$. Comparing $AG : GD$ with $AI : ID$ gives $x + y = a$, which is the claim. ■

With masses $1, 1, 1$ the same proof shows that a line through the centroid satisfies $\frac{PB}{PA} + \frac{QC}{QA} = 1$.

Problem 6

Let ABC be a scalene triangle whose incircle touches BC, CA, AB at D, E, F. Line EF meets line BC at X, line FD meets line CA at Y, and line DE meets line AB at Z. Prove that the midpoints of DX, EY and FZ are collinear.

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Since $AB \neq AC$, line EF, which is perpendicular to AI, is not parallel to BC, so X exists, and so do Y and Z. Menelaus for the line EFX gives $\frac{AF}{FB} \cdot \frac{BX}{XC} \cdot \frac{CE}{EA} = 1$, and the equal tangents $AF = AE$, $BF = BD$, $CE = CD$ turn this into

$$\frac{XB}{XC} = \frac{DB}{DC}.$$

As E and F lie inside two sides, X lies outside segment BC, while D lies inside that segment.

Let K be the midpoint of DX. Take coordinates on line BC with $K = 0$, $D = d$, $X = -d$, $B = \beta$, $C = \gamma$. Taking the positions of X and D into account, the displayed equation reads

$$\frac{\beta + d}{\gamma + d} = -\frac{\beta - d}{\gamma - d},$$

which simplifies to $\beta\gamma = d^2$, that is $\overline{KB} \cdot \overline{KC} = KD^2$.

Now KD^2 is the power of K with respect to the incircle, since KD is tangent to the incircle, and $\overline{KB} \cdot \overline{KC}$ is the power of K with respect to the circumcircle. So K lies on the radical axis of these two circles. This is a line, because a triangle whose incentre is its circumcentre is equilateral. The same argument puts the midpoints of EY and FZ on the same line. ■

Three points defined in the same way and claimed to be collinear are often on a radical axis, and the work is in finding the two circles.

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